Lakshya JEE 2.0 2025

Maths

Relations and Functions

DPP: 10

Q1 If $f: R \to R, f(x) = (x+1)^2$ and $g: R \to R, g(x) = x^2 + 1$, then $fog(-3)$ is equal to:

(A) 121 (B) 144

(C) 112 (D) 11

Q2 $f(x) = \sin^2 x + \sin^2\left(x + \frac{\pi}{3}\right) + \cos x \cos\left(x + \frac{\pi}{3}\right)$ and $g\left(\frac{5}{4}\right) = 1$, then $(g \circ f)(x)$ is equal to

(A) 1 (B) -1

(C) 2 (D) -2

Q3 If $g(x) = x^2 + x - 2$ and $\frac{1}{2}(gof)(x) = 2x^2 - 5x + 2$, then $f(x)$ is equal to:

(A) $2x - 3$

(B) $2x + 3$

(C) $2x^2 + 3x + 1$

(D) $2x^2 - 3x - 1$

Q4 If $f(y) = \frac{y}{\sqrt{1-y^2}}, g(y) = \frac{y}{\sqrt{1+y^2}}$, then $(fog)(y)$ is equal to:

(A) $\frac{y}{\sqrt{1-y^2}}$

(B) $\frac{y}{\sqrt{1+y^2}}$

(C) y

(D) $\frac{1-y^2}{\sqrt{1+y^2}}$

Q5 If $f(x) = \frac{2x-3}{x-2}$ then $[f\{f(x)\}]$ equals:

(A) x

(B) $-x$

(C) $\frac{x}{2}$

(D) $-\frac{1}{x}$

Q6 If $f(x) = \frac{1-x}{1+x'} x \neq 0, -1$ and $\alpha = f(f(x)) + f\left(f\left(\frac{1}{x}\right)\right)$, then

(A) $\alpha > 2$

(B) $\alpha < -2$

(C) $|\alpha| > 2$

(D) $\alpha = 2$

Q7 Let $f(x) = \frac{\alpha x}{x+1}, x \neq -1$. Then, for what value of α is $f(f(x)) = x$?

(A) $\sqrt{2}$

(B) $-\sqrt{2}$

(C) 1

(D) -1

Q8 $f: R \to R$ is a function defined by $f(x) = 10x - 7$. If $g = f^{-1}$, then $g(x) =$

(A) $\frac{1}{10x-7}$

(B) $\frac{1}{10x+7}$

(C) $\frac{x+7}{10}$

(D) $\frac{x-7}{10}$

Q9 Let $f(x) = x^2$ and $g(x) = 2^x$. Then, the solution set of the equation fog $(x) = g \circ f(x)$ is

(A) R (B) $\{0\}$

(C) $\{0, 2\}$ (D) None of these

Q10 If $f(x) = x^2 + 1$, then $f^{-1}(17)$ and $f^{-1}(-3)$ will be:

(A) $4, 1$ (B) $4, 0$

(C) $3, 2$ (D) None of these

Q11 If $f: [1, \infty) \to [1, \infty)$ is defined as $f(x) = 2^{x(x-1)}$ then $f^{-1}(x)$ is equal to:

(A) $\left(\frac{1}{2}\right)^{x(x-1)}$
(B) $\frac{1}{2}\left(1+\sqrt{1+4\log_2 x}\right)$
(C) $\frac{1}{2}\left(1-\sqrt{1+4\log_2 x}\right)$
(D) Not define

Q12 If $f:[1,\infty)\to[2,\infty)$ is given by $f(x)=x+\frac{1}{x}$ then $f^{-1}(x)$ equals:
(A) $\frac{x+\sqrt{x^2-4}}{2}$
(B) $\frac{x}{1+x^2}$
(C) $\frac{x-\sqrt{x^2-4}}{2}$
(D) $1+\sqrt{x^2-4}$

Q13 Let $f:[2,4)\to[1,3)$ be a function defined by $f(x)=x-\left[\frac{x}{2}\right]$ (where [.] denotes the greatest integer function). Then f^{-1} equals:
(A) x
(B) $x+1$
(C) $x+\left[\frac{x}{2}\right]$
(D) $x+2$

Q14 Values of 'x' satisfying both inequalities: $x-3\le 4 \& \frac{x-5}{x+1}>0$
(A) $x\in(-\infty,7]$
(B) $x\in(-\infty,-1)$
(C) $x\in(-\infty,-1)\cup(5,7]$
(D) $x\in(5,7]$

Q15 The inverse of the function $f:R\to\{x\in R:x<1\}$ given by $f(x)=\frac{e^x-e^{-x}}{e^x+e^{-x}}$, is
(A) $\frac{1}{2}\log\frac{1+x}{1-x}$
(B) $\frac{1}{2}\log\frac{2+x}{2-x}$
(C) $\frac{1}{2}\log\frac{1-x}{1+x}$
(D) None of these

Q16 Let $f:[4,\infty)\to[1,\infty)$ be defined by $f(x)=5^{x(x-4)}$. Then, $f^{-1}(x)$ is
(A) $\left(\frac{1}{5}\right)^{x(x-4)}$
(B) $2+\sqrt{4+\log_5 x}$
(C) $\left(\frac{1}{5}\right)^{x(x-4)}$
(D) None of these

Q17 If $F:[1,\infty)\to[2,\infty)$ is given by $f(x)=x+1/x$, then $f^{-1}(x)$ equals.
(A) $\frac{x+\sqrt{x^2-4}}{2}$
(B) $\frac{x}{1+x^2}$
(C) $\frac{x-\sqrt{x^2-4}}{2}$
(D) $1+\sqrt{x^2-4}$

Answer Key

Q1	(A)	Q10	(D)
Q2	(A)	Q11	(B)
Q3	(A)	Q12	(A)
Q4	(C)	Q13	(B)
Q5	(A)	Q14	(C)
Q6	(C)	Q15	(A)
Q7	(D)	Q16	(B)
Q8	(C)	Q17	(A)
Q9	(C)		

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